

SciTech Portfolio

Instruction Package



Introduction

A portfolio is a collection of your work that showcases you and your accomplishments. Many people in a variety of careers create portfolios and use them to secure jobs, gain promotions, bid for projects and document their professional growth.

In the SciTech program, you will gain experiences and certifications that go beyond the criteria required to earn the Ontario Secondary School Diploma. As a required component to earn the SciTech Certificate, the portfolio will enable you to encapsulate your skills, knowledge, and experiences in a professional document. Remember that the portfolio is your property and you can enhance the appearance and format to reflect your own individual strengths and skills.

Some portfolio items are self-selected and there are also some required components. Required components and artifacts are described in the following pages.

Your portfolio should be peer-reviewed on a yearly basis. In your graduating year, you will be presenting your full portfolio to staff. Presentation of your portfolio is a requirement to receive your SciTech Certificate.

Timelines and Required Components each Year

It will be important to work on your portfolio throughout the year. Lists of the items that must be included in your portfolio are outlined in the following checklist tables. To receive credit for required items, evidence must be included in the portfolio. Examples of evidence include photographs of completed work which include your photograph, rubrics and/or artifacts, such as completed projects. If no other evidence is available, *SciTech Activity Postcards* (p. 3) signed by a supervisor may be accepted, up to a maximum of one per school year. Field trips and events organized for all SciTech students DO NOT count towards extra-curricular experiences. **You will also need to complete reflections that include your comments:** What did you enjoy the most? What was most challenging? Why did you select the project? What insights did you take?

Please also take note of a few additional important points:

- The criteria "meets expectations" or "exceeds expectations" must be achieved on all components as described in the *Portfolio Assessment Rubric* (p. 8) in order to successfully earn the SciTech Certificate.
- Any part of the portfolio may be evaluated for marks during the four years of the SciTech Program as part of regular course work.
- Keep the following checklists (available online) as a permanent record of your accomplishments. They must be included in your portfolio each year.
- Academic Integrity - Your SciTech certificate may be placed in jeopardy should you claim others work as evidence of your own

SciTech Activity Postcard

Activity: at Secondary School

Contribution (hours, role, level of participation):

SciTech Activity Postcard

Activity: at Secondary School

Contribution (hours, role, level of participation):

Portfolio Components



PORT CREDIT
REGIONAL SCITECH PROGRAM

Grade 9	Self/Peer Check
Binder with dividers or a well-organized Digital Portfolio	☐
<i>SciTech Portfolio</i> Instruction Package	☐
<i>Portfolio Assessment Rubric</i>	☐
<u>Section #1: Personal Information</u>	
Why I Took SciTech	☐
My Future Goals (post secondary/career)	☐
Evidence for Academic Achievement (report cards, certificates)	☐
<u>Section #2: SciTech Skills & Knowledge & Certifications</u>	
Evidence of at least one completed science project (student-designed experiment or investigation, research project, etc.) including reflections	☐
Evidence of at least one completed technology project with a completed reflection for each.	☐
Evidence of Safe Practices in science <u>or</u> technology classrooms (tests, quizzes, certificates)	☐
<u>Section #3: Community Work & Extracurricular Involvement</u>	
At least one unique item involving community work and/or extracurricular involvement (i.e. volunteering, school clubs, councils) with completed reflection(s).	☐
<u>Section #4: Extracurricular SciTech Experiences</u>	
At least one unique item about your SciTech experiences (i.e. SciTech Council, Coding Club, Regional Science Fair, Skills Canada, Rocketry, Robotics, Junior Solar Car, contests and competitions) with completed reflection(s).	☐



Grade 10

Self-reflection

- *Portfolio Assessment Rubric*

☐

Section #1: Personal Information

- Evidence for Academic Achievement (report cards, certificates)
- Career investigation summary (e.g. *Career Cruising, Ontario Skills Passport, MyBlueprint, Making My Way, etc.*)
- Updated Resume and Cover Letter

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Section #2: SciTech Skills & Knowledge & Certifications

- Evidence of at least one completed science project (student-designed experiment or investigation, research project) including reflections
- Evidence of at least one completed technology project including reflections
- Evidence of Safe Practices in science or technology classrooms (tests, quizzes, certificates)

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Section #3: Community Work & Extracurricular Involvement

- At least one unique, organized item involving community work and/or extracurricular involvement that shows evidence of growth from the previous grade. Include a completed reflection(s)

☐

Section #4: Extracurricular SciTech Experiences

- At least one unique item about your SciTech experiences that shows evidence of growth from the previous grade. Include a completed reflection(s)

☐

*See Appendix for a list of appropriate items



Grade 11

Self/Peer Check

- *Portfolio Assessment Rubric*

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Section #1: Personal Information

- Post-Secondary and Career Plans
- Evidence for Academic Achievement (report cards, certificates)

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Section #2: SciTech Skills & Knowledge & Certifications

- Evidence of participation in a STEM related contest organized at PCSS
- Evidence of at least one completed science project (student-designed experiment or investigation, research project) including reflections
- Evidence of at least one completed technology projects including reflections

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Section #3: Community Work & Extracurricular Involvement

- At least one unique, organized item involving community work and/or extracurricular involvement that shows evidence of growth from the previous grade. Include a completed reflection(s)

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Section #4: Extracurricular SciTech Experiences

- At least one unique item about your SciTech experiences that shows evidence of growth from the previous grade. Include a completed reflection(s)

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*See Appendix for a list of appropriate items



Grade 12

Self/Peer Check

- *Portfolio Assessment Rubric*

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Section #1: Personal Information

- Updated Resume & Cover Letter showcasing Employability/Learning Skills
- Evidence for Academic Achievement (report cards, certificates)

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Section #2: SciTech Skills & Knowledge & Certifications

- Evidence of participation in a STEM related contest organized at PCSS
- Evidence of at least one completed science project (student-designed experiment or investigation, research project) including reflection
- Evidence of at least one completed technology project including reflections

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Section #3: Community Work & Extracurricular Involvement

- At least one unique, organized item involving community work and/or extracurricular involvement that shows evidence of growth from the previous grade. Include a completed reflection(s)

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Section #4: Extracurricular SciTech Experiences

- At least one unique item about your SciTech experiences that shows evidence of growth from the previous grade. Include a completed reflection(s)

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**See Appendix for a list of appropriate items*

Portfolio Assessment Rubric



Student Name: _____

Year: ☐ 1 ☐ 2 ☐ 3 ☐ 4

Reviewer: _____

Date: _____

	Exceeds Expectations	Meets Expectations	Significant Gaps - Resubmit
Completion	All mandatory items as well as additional components completed fully within the timelines provided	All mandatory items completed fully within the timelines provided	Items missing or incomplete
Item Selection	Items chosen amply demonstrate accomplishments, skills, and growth	Items chosen demonstrate accomplishments, skills, and growth	Weak demonstration of accomplishments; viewer is left with many questions unanswered
Organization	Clear, easy-to-follow, well-organized. Sub-sections are used and a separate table of contents guides the reader	Clear, easy-to-follow, well-organized.	Confusing, unclear, poorly organized; too much effort required to view it
Critical Thinking	Insightful reflections exhibit depth of understanding, sincerity and serious focus	Reflections exhibit understanding, sincerity and serious focus	Superficial treatment of reflections leads reader to feel student is insincere or unfocused
Communication	Grammar and writing conventions correct; captures reader's attention through effective use of language	Grammar and writing conventions correct; clear use of language	Errors in grammar and writing conventions indicate minimal editing or proofreading
Overall Impression	Professional in all ways - a true keepsake for future use	Professional document for future use	Rush job with little attention to detail

Comments:



Appendix

See appropriate examples of items you may choose to include and reflect upon in your SciTech Portfolio. If unsure, please see your SciTech teacher for guidance.

STEM Related Contests:

- *Caribou Math Contests*
- *Gauss Mathematic Contests*
- *Canadian Senior and Intermediate Mathematics Contests*
- *Canadian Lynx Mathematical Competition*
- *Pascal, Cayley, and Fermat Mathematics*
- *LinkedKey Mathematics Program*
- *Avogadro's Exam*
- *Chem 13 News Exam*
- *Sir Isaac Newton Physics Exam*
- Formal Science Fair
- HOSA exam or competition

Community Work & Extracurricular Involvement:

- volunteering and/or participating in a formal event for a formal and recognized organization with signed proof from a searchable/contactable member of said organization (it does NOT need to be Science or Tech related)

****cannot be an independent project for yourself or paid work experience

Extracurricular SciTech Experiences:

- *Robotics Club*
- *Astronomy Club*
- Volunteering at a hospital or medical/veterinary clinic
- Coding Club or courses
- *Stage Crew*
- *HOSA Club*
- *Pre-Med Club*
- *SciTech Student Council*

****must be done outside of class time and can be an extracurricular activity done outside of PCSS but must be organized by a formal organization with signed proof from a searchable/contactable member of said organization (it DOES need to be obviously Science or Technology related)